FIELD GUIDE · AOR / GE(N) MAINT ISLD

Power Monitor Field Guide

Everything on the dashboard, in plain words: what each term means, what each tab shows, and — most importantly — which numbers come from your asset register and which are simulated stand-ins waiting for real meters.

110 register points **107** monitored **12** tabs **3** substations **109** transformers **2,157** buildings

it works Power terms Network terms Power quality SCADA terms Alarm terms Analysis terms

START HERE

The one thing to understand first

Your source document is an *asset register* — it lists what exists and how many. It contains no meter readings of any kind. So every figure on screen carries a small badge saying where it came from.

REG

From the register

Counted or stated in the source document. These are facts you can defend.

- Asset counts — 2,157 buildings, 980 street lights, 84 pumps
- Transformer fleet — 109 units, 18,530 kVA installed
- Named locations — tube wells, filter plants, water tanks
- Network shape — 3 substations, 3 feeders, 8 breakers, 475 boards

SIM

Simulated

Modelled stand-ins, because no meters exist yet. Realistic and internally consistent, but *not measurements*.

- Every live reading — kW, voltage, current, frequency
- Power factor, reactive power, energy totals
- Breaker open/closed positions

- All alarm limits (they derive from estimated ratings)

> **How to say it in one sentence.** “The structure is real and traceable to our register; the live numbers are a working simulation, clearly marked, and they get replaced by the real meter feed without changing anything you see.”

THE CHAIN

How a reading becomes an alarm

Four steps. Understanding this explains almost every behaviour on the dashboard.

STEP 1

A meter reports

Each location sends its power draw with a timestamp, every 2 seconds. Today this comes from the simulator; later it comes from your SCADA system.

STEP 2

The reading is smoothed

Raw sensor readings jitter. A 6-second average is applied so normal noise doesn’t look like the plant misbehaving. The raw value is still shown in the detail panel.

STEP 3

It’s checked against limits

Every location has a warning limit and an alarm limit. The reading is compared against both, plus its typical value for this hour.

STEP 4

An alarm is raised — carefully

Only if the breach lasts 6 seconds, and only once per event. It then stays on screen until someone acknowledges it.

GLOSSARY · 01

Power and energy

The difference between power and energy trips people up more than anything else here.

Power

`kW` · `MW`

How fast electricity is being used *right now* — like the speedometer in a car. 1 MW = 1,000 kW. A kettle is about 2 kW; the whole complex runs around 4,500 kW.

Energy

`kWh` · `MWh`

How much electricity has been *used over time* — like the odometer. This is what the electricity bill charges for. Running 1 kW for 1 hour = 1 kWh.

Active power

`kW`

The useful part of the power — what actually turns motors and produces light and heat. Also called *real power*.

Reactive power

`kVAR · MVAR`

Power that sloshes back and forth without doing useful work. Motors, pumps and transformers all need it to build their magnetic fields. It does no work but it still occupies capacity in the cables.

`On the dashboard: the second KPI tile.`

Apparent power

`kVA · MVA`

Active and reactive power combined — the total the cables and transformers must physically carry. This is why transformers are rated in kVA, not kW.

Demand

The power being drawn at a given moment. *Peak demand* is the highest it reached. Utilities often charge extra based on peak demand, so it matters commercially.

Connected load

The total of everything that *could* be switched on at once. Always much larger than actual demand, because not everything runs simultaneously.

`Site connected capacity: 16.7 MW. Typical demand: ~4.5 MW.`

Diversity

The reason for that gap. In 2,157 buildings, people don't all boil kettles at the same second, so the real total is far below the theoretical maximum.

GLOSSARY · 02

The electrical network

Electricity arrives at high voltage and is stepped down in stages until it reaches a socket. These are the pieces of that chain.

High tension (HT)

`11 kV`

High-voltage distribution. Electricity travels at high voltage because it loses less energy that way. Your site distributes at 11,000 volts.

`Register states “VCBs HT 11 KV — 8 Nos”, and 38 km of HT cable.`

Low tension (LT)

`400 V`

The voltage after stepping down — what actually reaches buildings and sockets.

`108 km of LT cable on site.`

Substation

Where high voltage is received and stepped down for local distribution.

`Three: PNS Hafeez, ARC, NHQ — plus 13 satellite stations for outstations.`

Transformer

`kVA`

The device that changes voltage from high to low. Rated in kVA — its maximum throughput. A 200 kVA unit can carry roughly 180 kW of real load.

`109 units totalling 18,530 kVA: 1×1250, 1×630, 73×200, 20×100, 1×50, plus 13 satellite.`

Feeder

A cable circuit carrying power from a substation out to a group of loads. Think of it as a trunk road for electricity.

`Register states 3 feeders — one per substation.`

Busbar (bus)

A heavy metal bar inside a substation that everything connects to — the common junction where incoming supply meets the outgoing feeders.

Circuit breaker

`VCB`

An automatic switch that disconnects a circuit when something goes wrong. A *Vacuum Circuit Breaker* interrupts the current inside a vacuum chamber, which stops the arc cleanly. Used at high voltage.

`8 VCBs at 11 kV across TS Rawat, PNS Hafeez, ARC and OM.`

Distribution board

`DB`

The panel of fuses or breakers inside a building that splits incoming supply into individual circuits — the consumer unit in a house, scaled up.

`475 boards. The register gives only the total, no names or locations.`

Single line diagram

`SLD`

The standard engineering drawing of an electrical network, showing supply → bus → substations → loads as one simplified line per circuit. Every electrician reads these.

Solar PV

`kWp`

Panels that generate electricity from sunlight, reducing what must be imported. *kWp* is “kilowatt peak” — output in full sun.

`13 systems. The register gives the count but no ratings.`

GLOSSARY · 03

Power quality

Not just *how much* electricity, but *how good* it is. Poor quality damages equipment and wastes money even when supply never fails.

Voltage

`V · kV`

The electrical “pressure” pushing current through the wires. Too low and motors overheat; too high and insulation degrades.

`Nominal 11 kV, allowed range 10.5–11.5 kV.`

Line-to-line

`L-L`

Voltage measured between two of the three phases, rather than between one phase and earth. It’s the standard way to quote three-phase voltage.

Current

`A (amps)`

How much electricity is flowing. Cables are sized by current — exceed their rating and they overheat.

Frequency

`Hz`

How many times per second the current alternates direction. It must stay very close to 50 Hz — it is a live indicator of whether national generation matches national demand.

`Statutory band 49.5–50.5 Hz.`

Three phase

`R-Y · Y-B · B-R`

Industrial supply uses three separate live conductors — Red, Yellow, Blue — offset in timing so power delivery is smooth and motors run efficiently. The three labels are the three pairings between them.

Phase imbalance

When the three phases carry noticeably different loads. This overheats motors and wastes energy. Should stay under 2%.

`Headlined on the Voltage & Frequency tab because it causes real damage.`

Power factor

`PF, 0–1`

How efficiently power is used — the share of supplied power doing useful work. 1.0 is perfect; 0.95 is good; below 0.90 usually attracts a utility penalty. Lots of motors pulls it down. Improving it is often the cheapest saving available.

Lagging

Describes the usual direction of power-factor error, caused by motors and transformers. Normal on any industrial site — not a fault by itself.

Distribution losses

Electricity lost as heat in cables and transformers between intake and end use. Always present; typically 2–4%. You pay for it but never use it.

GLOSSARY · 04

Monitoring and data

How readings get from a meter to the screen, and how the dashboard judges whether it can trust them.

SCADA

Supervisory Control and Data Acquisition — the standard family of industrial systems that collect readings from field equipment and present them centrally. This dashboard is designed to sit on top of one.

Telemetry

Measurements sent automatically from remote equipment. The opposite of someone walking round with a clipboard.

Tag / Point ID

`TW-01, MED-HOSP`

The unique code identifying one monitored location. Everything keys off it — readings, limits, alarms and history.

`TW-01 is Tube Well No.1; MED-HOSP is PNS Hafeez Hospital.`

Polling vs streaming

`REST · SSE`

Two ways to get data. *Polling* means the dashboard asks “anything new?” every few seconds. *Streaming* means the server pushes updates the instant they happen — faster and lighter. Both are supported; switch in the header.

Smoothing

`EMA, 6 s`

Averaging out momentary jitter so a stable plant looks stable. *Exponential Moving Average* weights recent readings more heavily than older ones.

Data quality

Whether a reading can be trusted — deliberately kept separate from whether the value is high or low. A meter that has gone silent is a completely different problem from a meter correctly reporting zero, and the dashboard never confuses the two.

Stale

`> 15 s`

No fresh reading recently. The value shown is real but slightly old.

Offline

`> 40 s`

The meter has stopped reporting entirely — a communications fault. Shown distinctly, never as “0 kW”.

Watchdog

A background check that watches for *silence* rather than errors. If the whole feed dies, a banner says so loudly instead of the screen quietly freezing on old numbers — the most dangerous failure a dashboard can have.

GLOSSARY · 05

Alarm handling

These follow ISA-18.2, the international standard for industrial alarms. The rules exist because naive alarming — beep whenever a number is high — floods operators until they stop looking.

Warning limit

The early-notice threshold. Something is drifting; worth a look, not urgent.

Alarm limit

The real limit. Consumption has exceeded what this location should draw and needs attention.

On-delay

`6 s`

A limit must be exceeded *continuously* for 6 seconds before an alarm raises. One noisy sample can't trip it.

Hysteresis / deadband

4%

To clear, a value must fall 4% *below* the limit — not just back to it. Without this, a value hovering exactly on the threshold would alarm and clear repeatedly, endlessly. Same idea as a thermostat.

Off-delay

`12 s`

It must also stay clear for 12 seconds before the alarm is considered over.

Latching

An alarm stays visible until a person acknowledges it, even if the problem fixes itself. Nothing ever disappears unseen.

Acknowledge (Ack)

An operator pressing "Ack" to confirm they've seen it. The time is recorded in the log.

Ringback

An alarm that fixed itself *before* anyone acknowledged it stays on the list anyway, marked cleared. Otherwise intermittent faults would go permanently unnoticed.

Escalation

An existing alarm getting worse — warning becoming critical. It updates the same alarm rather than raising a second one, and cancels any prior acknowledgement.

Re-arm

`90 s`

After an alarm closes, the same location can't raise the same alarm again for 90 seconds. Stops one flickering sensor generating dozens of entries.

`The point still displays as over limit — only the new alarm is held back.`

Alarm flood

The failure this all guards against: so many alarms at once that operators stop reading them. The standard defines it as more than 10 in 10 minutes.

Episode

One complete alarm event, from raised through acknowledged to cleared, recorded as a single entry with its peak value — not scattered across dozens of rows.

GLOSSARY · 06

Analysis terms

Terms that appear on the Energy, Solar and Forecasting tabs.

Typical for this hour

What this location normally draws at this time of day. Every reading is shown against it, because "640 kW" means nothing until you know whether that's normal.

Load profile

The shape of demand across a day. Housing peaks in the evening; offices peak midday; street lighting only runs at night. Each asset group has its own.

Load factor

Average demand ÷ peak demand, as a percentage. High means steady, efficient use. Low means you're paying for capacity to cover brief spikes.

Load duration curve

Demand sorted highest to lowest instead of by time. It answers a question a normal trend can't: *how much of the time are we actually near peak?* A steep drop at the left means peaks are brief and rare.

Headroom

Spare capacity between forecast peak demand and what the network can safely carry. This is what tells you whether a new building can be connected.

Firm capacity

The load the network can carry safely and continuously — based on installed transformer capacity.

Performance ratio

For solar: actual output ÷ what should be produced given the current sun angle. This catches a dirty or faulty panel, which a raw output figure never would — low output at dawn is normal, low output at noon is a fault.

Emission factor

`0.42 kg/kWh`

CO_2 produced per unit of grid electricity. Multiplying by solar generation gives the carbon avoided.

Confidence range

The shaded band on the forecast. It widens further into the future because predictions get less certain — an honest forecast shows its own uncertainty.

EVERYWHERE

Elements shared by every tab

These appear on all screens, so explain them once.

The top bar

Clock and date

Current time. Confirms the screen is live, not a frozen screenshot.

LIVE indicator

Pulsing green dot when data is flowing. Dims when paused or disconnected.

Connection pill

Feed health and age of the last message — “Live · SSE · 2s ago”. Turns amber if delayed, red if lost.

Source selector

Switch between the built-in simulator, a live SCADA stream, or REST polling. This is where you point it at real data.

Pause button

Freezes updates so you can read or discuss a value without it moving.

Alarm bell

Count of unacknowledged alarms. Click to jump to the alarm list.

A — wall display

Enlarges everything for a screen read from across a control room.

Sun / moon

Light and dark themes. Dark suits control rooms; light suits printing and projectors.

Status colours and badges

Every status carries an icon and a word as well as a colour, so it still reads correctly for colourblind viewers and in print.

Normal	Within limits. Nothing to do.
Warning	Above the warning limit. Drifting — worth a look.
Over limit	Above the alarm limit. Needs attention now.
Offline	Meter not reporting. A communications problem, not a power problem.
Not monitored	In the register but has no meter — e.g. tube wells TW-02, 06, 12.

SCREEN BY SCREEN

The 12 tabs

In the order they appear down the left-hand navigation.

1 Overview

`the “is anything wrong” screen`

The one screen to leave on a wall. Everything needed to answer “is the site healthy right now?” without scrolling.

Six KPI tiles

Total Active Power (MW) · Reactive Power (MVAR) · Power Factor · Frequency (Hz) · Average Voltage (kV) · Energy Today (kWh). Each has a green tick or amber warning showing whether it’s in its normal band.

Total load chart

Site demand over time, with a dashed line showing typical for this hour. Hover for exact values at any point.

Active alarms

Anything currently over limit, worst first, each with an Ack button. Says “No points over limit” when all is well.

Power distribution

Doughnut splitting demand across the three substations plus solar, with kW and percentage for each.

Load by category

Bar chart of the biggest consuming asset groups. A bar turns amber or red if that group holds an alarm.

Energy consumption

Gauge of energy used this session against what would be expected for the time elapsed.

Energy summary

Consumed · solar generated · net import · distribution losses.

Single line diagram

The 11 kV network: grid → main bus → three substations, plus the solar infeed. Each substation shows its live load, current, power factor and a loading bar. Its box outlines amber or red if a load beneath it is in alarm.

Voltage profile

All three phases over time, with the 10.5 and 11.5 kV statutory limits drawn as red dashed lines.

Six metric tiles

CO_2 avoided · losses · efficiency · connected capacity · max demand · peak current.

Register data gaps

Discrepancies found while reconciling the source document — questions for the facilities team, listed openly rather than hidden.

LOOK AT The four numbers that matter: **alarm count, frequency, power factor and voltage**. If those are green, the site is fine.

2 Power Flow

`where the electricity goes`

Traces power from where it enters the site to where it's consumed. The width of each ribbon is proportional to the power flowing through it — thick means a lot, thin means a little.

Four stat tiles

Total demand · solar contribution · estimated losses · delivery efficiency.

Flow diagram

Four columns left to right: **Source** (grid and solar) → **Main 11 kV bus** → **Substation** → **Load group**. Follow any ribbon to see exactly where power ends up.

Feeder loading

How hard each of the three feeders is working, as a percentage of installed capacity. Amber above 80%, red above 90%.

`USE IT FOR` Explaining the site to someone unfamiliar with it, and spotting which group is driving a spike.

3 Load Monitoring

`every location, in detail`

The full working list — all 110 register locations, searchable and sortable. This is the tab an engineer investigating a problem will live in.

Load group cards

Nine cards, one per asset group, each with live kW, status and a loading bar. Click one to filter the table beneath.

Search and filters

Free-text search on tag, name or location; dropdowns for group and substation; and a "only points needing attention" toggle.

The table columns

Tag — the ID. **Location** — name and place. **Now** — live draw. **Typical** — normal for this hour. **vs typical** — percentage above or below. **Limit** — its alarm limit. **% of limit** — how close it is. **Trend** — a mini chart of recent history. **Status** — the badge. **Updated** — how long since the last reading.

Detail panel

Click any row: its trend with warning, alarm and rated lines drawn on it, plus 15 facts

(smoothed and raw values, peak, energy, unit count, feeder, service state) and that location's own alarm history.

SORTING Defaults to worst-first, so problems float to the top automatically. Click any column heading to re-sort.

4 Voltage & Frequency

`supply quality`

Whether the electricity arriving is *good*, separate from how much of it there is. Poor quality damages equipment even when nothing ever trips.

Four stat tiles

System frequency · average L-L voltage · worst phase imbalance · line current. Each says whether it's inside its statutory band.

Frequency chart

Frequency over time with the 49.5 and 50.5 Hz limits and the 50.0 Hz nominal marked.

Three voltage charts

One per substation, each showing all three phases against the 10.5–11.5 kV band.

Measurements table

Per substation: all three phase voltages, imbalance percentage, current, power factor and reactive power.

LOOK AT **Phase imbalance.** It's given its own tile because sustained imbalance quietly destroys motors — the most expensive failure on this list.

5 Energy Analytics

`consumption over time — the billing view`

Shifts from power (right now) to energy (accumulated). This is the tab for cost conversations rather than operational ones.

Four stat tiles

Energy consumed · average demand · peak demand and when it happened · load factor.

Energy mix

Doughnut splitting total energy into grid import, solar generated and distribution losses.

Load duration curve

Demand sorted high to low. Shows what proportion of time is spent near peak — a question the ordinary trend cannot answer.